

## SPIRIT OF ST LOUIS, MO, USA

WMO: 724345

Lat: 38.657N

Lon: 90.656W

Elev: 141

StdP: 99.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03966

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.7      | -11.5 | -20.1                         | 0.6 | -12.7 | -17.5 | 0.8 | -10.1 | 10.9                  | 1.3  | 9.6 | 1.2  | 2.9                   | 290  | 0.461 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.2                   | 35.2            | 25.2 | 33.8 | 24.7 | 32.5 | 24.1 | 26.6                | 32.6 | 25.8 | 31.6 | 25.1 | 30.7 | 3.4                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.0                            | 20.4 | 29.8 | 24.1 | 19.3 | 29.1 | 23.4 | 18.5 | 28.4 | 83.8          | 32.7 | 80.5 | 31.7 | 77.3 | 30.8 | 29.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|-----|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |     | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c) | (d)                        | (e) | (f)                | (g)  | (h)                                                | (i) | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  | (p)   | (q)  |
| (4)               | 9.2  | 8.2 | 7.3                        | DB  | -19.2              | 37.7 | 3.3                                                | 1.9 | -21.6      | 39.1 | -23.5      | 40.2 | -25.4      | 41.2 | -27.8 | 42.6 |
| (5)               |      |     |                            | WB  | -19.6              | 27.9 | 3.2                                                | 0.9 | -21.9      | 28.5 | -23.8      | 29.0 | -25.6      | 29.5 | -28.0 | 30.1 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 13.5   | -0.3  | 2.2   | 7.7   | 13.6  | 18.9  | 23.7  | 25.8  | 25.0  | 20.5  | 14.1  | 7.5   | 2.0  |
|                                                                                  | DBStd       | 10.37  | 6.58  | 6.16  | 6.08  | 5.23  | 4.49  | 3.38  | 3.11  | 3.05  | 4.17  | 5.07  | 5.43  | 5.73 |
|                                                                                  | HDD10.0     | 1078   | 324   | 229   | 117   | 22    | 1     | 0     | 0     | 0     | 0     | 17    | 111   | 256  |
|                                                                                  | HDD18.3     | 2584   | 578   | 453   | 332   | 159   | 50    | 3     | 0     | 1     | 25    | 149   | 326   | 508  |
|                                                                                  | CDD10.0     | 2337   | 5     | 10    | 47    | 129   | 277   | 412   | 490   | 465   | 315   | 145   | 36    | 7    |
|                                                                                  | CDD18.3     | 803    | 0     | 0     | 3     | 16    | 68    | 166   | 232   | 207   | 90    | 19    | 1     | 0    |
|                                                                                  | CDH23.3     | 8284   | 0     | 2     | 40    | 194   | 620   | 1656  | 2561  | 2088  | 904   | 204   | 14    | 0    |
|                                                                                  | CDH26.7     | 3444   | 0     | 0     | 4     | 47    | 188   | 684   | 1202  | 926   | 340   | 51    | 1     | 0    |
| Wind                                                                             | WSAvg       | 2.9    | 3.4   | 3.5   | 3.6   | 3.6   | 2.9   | 2.5   | 2.2   | 2.0   | 2.0   | 2.6   | 3.1   | 3.1  |
| Precipitation                                                                    | PrecAvg     | 1098   | 65    | 57    | 84    | 119   | 133   | 119   | 97    | 89    | 90    | 82    | 90    | 71   |
|                                                                                  | PrecMax     | 1552   | 220   | 116   | 178   | 295   | 301   | 254   | 194   | 158   | 306   | 348   | 196   | 260  |
|                                                                                  | PrecMin     | 688    | 17    | 10    | 35    | 45    | 34    | 27    | 27    | 36    | 15    | 20    | 22    | 18   |
|                                                                                  | PrecStd     | 199    | 46    | 30    | 34    | 55    | 71    | 59    | 45    | 35    | 64    | 63    | 54    | 50   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 20.2  | 22.5  | 27.1  | 30.8  | 32.5  | 35.4  | 38.2  | 37.2  | 34.2  | 30.8  | 25.4  | 20.7 |
|                                                                                  |             | MCWB   | 14.8  | 14.6  | 17.1  | 19.4  | 22.6  | 24.5  | 24.7  | 24.9  | 23.4  | 21.8  | 17.9  | 16.2 |
|                                                                                  | 2%          | DB     | 16.3  | 19.0  | 24.2  | 27.9  | 30.7  | 33.7  | 35.6  | 35.0  | 32.4  | 27.9  | 22.1  | 17.3 |
|                                                                                  |             | MCWB   | 12.3  | 13.0  | 15.9  | 17.7  | 21.8  | 24.5  | 25.7  | 25.0  | 23.3  | 19.8  | 15.0  | 13.6 |
|                                                                                  | 5%          | DB     | 12.9  | 16.1  | 21.3  | 25.7  | 28.8  | 32.3  | 33.9  | 33.1  | 30.6  | 25.7  | 19.3  | 14.0 |
|                                                                                  |             | MCWB   | 8.8   | 11.1  | 14.1  | 17.0  | 21.1  | 24.1  | 25.2  | 24.4  | 22.3  | 18.4  | 14.3  | 10.1 |
|                                                                                  | 10%         | DB     | 9.6   | 12.6  | 18.3  | 23.0  | 27.1  | 30.8  | 32.5  | 31.5  | 28.4  | 23.1  | 17.2  | 11.2 |
|                                                                                  |             | MCWB   | 6.2   | 8.4   | 12.7  | 15.8  | 20.1  | 23.2  | 24.6  | 23.6  | 21.1  | 17.3  | 12.8  | 7.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.9  | 16.4  | 18.8  | 20.7  | 24.4  | 26.7  | 28.0  | 27.3  | 25.4  | 23.1  | 18.8  | 16.9 |
|                                                                                  |             | MCDB   | 19.4  | 20.2  | 25.4  | 27.4  | 30.0  | 32.4  | 33.5  | 33.2  | 31.6  | 28.5  | 22.7  | 19.9 |
|                                                                                  | 2%          | WB     | 12.9  | 14.4  | 16.9  | 19.3  | 23.2  | 25.6  | 26.9  | 26.3  | 24.4  | 21.5  | 16.8  | 14.1 |
|                                                                                  |             | MCDB   | 15.7  | 18.2  | 22.2  | 25.3  | 28.8  | 31.6  | 32.8  | 32.4  | 30.4  | 25.9  | 20.1  | 16.8 |
|                                                                                  | 5%          | WB     | 9.5   | 11.5  | 15.4  | 18.2  | 22.1  | 24.9  | 26.1  | 25.5  | 23.5  | 19.9  | 15.3  | 10.9 |
|                                                                                  |             | MCDB   | 12.8  | 15.3  | 20.0  | 23.6  | 27.1  | 30.6  | 32.1  | 31.2  | 28.5  | 23.8  | 18.7  | 13.1 |
|                                                                                  | 10%         | WB     | 6.2   | 8.1   | 13.4  | 16.9  | 21.1  | 24.0  | 25.4  | 24.7  | 22.5  | 18.2  | 13.0  | 8.0  |
|                                                                                  |             | MCDB   | 9.5   | 12.4  | 18.0  | 21.9  | 25.7  | 29.3  | 31.0  | 29.9  | 26.7  | 22.3  | 16.9  | 10.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.0  | 10.5  | 11.9  | 12.5  | 11.5  | 11.1  | 11.2  | 11.6  | 12.8  | 12.7  | 11.5  | 9.8  |
|                                                                                  |             | MCDBR  | 15.4  | 16.4  | 15.9  | 15.9  | 13.2  | 12.7  | 12.5  | 13.2  | 14.4  | 15.3  | 15.4  | 14.2 |
|                                                                                  |             | MCWBR  | 10.9  | 11.1  | 8.9   | 8.2   | 6.2   | 5.4   | 4.8   | 5.0   | 6.1   | 7.9   | 9.9   | 10.4 |
|                                                                                  | 5% WB       | MCDBR  | 13.4  | 14.7  | 13.5  | 13.1  | 11.4  | 11.4  | 11.3  | 11.5  | 11.9  | 12.0  | 13.1  | 12.9 |
|                                                                                  |             | MCWBR  | 10.5  | 11.0  | 8.7   | 7.7   | 5.9   | 5.4   | 5.0   | 4.9   | 5.4   | 7.2   | 9.7   | 10.6 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.304  | 0.318 | 0.345 | 0.399 | 0.439 | 0.449 | 0.475 | 0.466 | 0.408 | 0.352 | 0.311 | 0.303 |      |
|                                                                                  | taud        | 2.477  | 2.428 | 2.419 | 2.263 | 2.211 | 2.220 | 2.148 | 2.178 | 2.312 | 2.475 | 2.531 | 2.496 |      |
|                                                                                  | Ebn at Noon | 877    | 909   | 915   | 878   | 846   | 835   | 809   | 806   | 836   | 859   | 863   | 851   |      |
|                                                                                  | Edh at Noon | 79     | 95    | 106   | 132   | 142   | 141   | 151   | 143   | 118   | 90    | 75    | 73    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.07   | 2.77  | 3.75  | 4.96  | 5.57  | 6.35  | 6.21  | 5.55  | 4.76  | 3.42  | 2.32  | 1.74  |      |
|                                                                                  | RadStd      | 0.23   | 0.28  | 0.36  | 0.29  | 0.49  | 0.49  | 0.43  | 0.32  | 0.30  | 0.38  | 0.24  | 0.18  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +83     | +82     |
| (47) | Regional (26 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +82     |

Nomenclature: See separate page